WORKED SOLUTION

STEP I 2012, Question 5

Pure • Integration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q5

The key idea

Both of the first two integrals give way to standard tools. Writing $\sin 2x = 2\sin x \cos x$ shows that $\sin x$ is (minus) the derivative of $\cos x$, so the substitution $u = \cos x$ works. For the second, the logarithm is the awkward factor, so we integrate by parts to differentiate it away. For the final integral, $\cos x + \sin x$ is a cosine in disguise: $\sqrt{2}\cos(x - \frac{1}{4}\pi)$. Shifting the variable by $\frac{1}{4}\pi$ turns the range into $[0, \frac{1}{4}\pi]$ and the integrand into a combination of the two integrals we have just found.

Solution

The first integral

Write I_1 for the first integral. Since $\sin 2x = 2\sin x \cos x$, substitute $u = \cos x$, so $\mathrm{d}u = -\sin x \,\mathrm{d}x$. When $x = 0$, $u = 1$, and when $x = \frac{1}{4}\pi$, $u = \frac{1}{\sqrt{2}}$. So

$$I_1 = \int_0^{\frac{1}{4}\pi} 2\cos x \ln(\cos x)\, \sin x \,\mathrm{d}x = \int_{1/\sqrt{2}}^{1} 2u \ln u \,\mathrm{d}u,$$

where the minus sign from $\mathrm{d}u$ has been used to swap the limits. Integrating by parts (differentiate $\ln u$, integrate $2u$),

$$\int 2u \ln u \,\mathrm{d}u = u^2 \ln u - \int u^2 \cdot \frac{1}{u} \,\mathrm{d}u = u^2 \ln u - \tfrac{1}{2}u^2.$$

At $u = 1$ this is $-\frac{1}{2}$. At $u = \frac{1}{\sqrt{2}}$ we have $u^2 = \frac{1}{2}$ and $\ln u = -\frac{1}{2}\ln 2$, so it is $-\frac{1}{4}\ln 2 - \frac{1}{4}$. Hence

$$I_1 = -\tfrac{1}{2} - \left(-\tfrac{1}{4}\ln 2 - \tfrac{1}{4}\right) = \tfrac{1}{4}\ln 2 - \tfrac{1}{4} = \tfrac{1}{4}(\ln 2 - 1),$$

as required.

The second integral

Write I_2 for the second integral. We integrate by parts, differentiating $\ln(\cos x)$ and integrating $\cos 2x$. A convenient antiderivative of $\cos 2x$ is $\frac{1}{2}\sin 2x = \sin x \cos x$, and the derivative of $\ln(\cos x)$ is $-\sin x / \cos x$. So

$$\begin{aligned} I_2 &= \Big[\sin x \cos x \, \ln(\cos x)\Big]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \sin x \cos x \cdot \left(-\frac{\sin x}{\cos x}\right) \mathrm{d}x \\ &= \tfrac{1}{2}\ln\left(\tfrac{1}{\sqrt{2}}\right) - 0 + \int_0^{\frac{1}{4}\pi} \sin^2 x \, \mathrm{d}x. \end{aligned}$$

Ask yourself Why choose $\sin x \cos x$ rather than just writing $\frac{1}{2}\sin 2x$?

They are the same function, but in the form $\sin x \cos x$ the factor $\cos x$ cancels with the $\cos x$ in the denominator of the derivative of $\ln(\cos x)$, leaving the easy integral of $\sin^2 x$.

The boundary term is $\frac{1}{2} \cdot (-\frac{1}{2}\ln 2) = -\frac{1}{4}\ln 2$. Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

$$\int_0^{\frac{1}{4}\pi} \sin^2 x \, \mathrm{d}x = \Big[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\Big]_0^{\frac{1}{4}\pi} = \tfrac{\pi}{8} - \tfrac{1}{4}.$$

So

$$I_2 = -\tfrac{1}{4}\ln 2 + \tfrac{\pi}{8} - \tfrac{1}{4} = \tfrac{1}{8}\left(\pi - 2\ln 2 - 2\right) = \tfrac{1}{8}(\pi - \ln 4 - 2),$$

as required.

The final integral

Call the final integral I. By the compound-angle formula,

$$\cos x + \sin x = \sqrt{2}\left(\cos x \cos \tfrac{1}{4}\pi + \sin x \sin \tfrac{1}{4}\pi\right) = \sqrt{2}\cos\left(x - \tfrac{1}{4}\pi\right).$$

This suggests the substitution $t = x - \frac{1}{4}\pi$, which sends the range $[\frac{1}{4}\pi, \frac{1}{2}\pi]$ to $[0, \frac{1}{4}\pi]$, the range of I_1 and I_2. With $2x = 2t + \frac{1}{2}\pi$,

$$\cos 2x = \cos\left(2t + \tfrac{1}{2}\pi\right) = -\sin 2t, \qquad \sin 2x = \sin\left(2t + \tfrac{1}{2}\pi\right) = \cos 2t,$$

and

$$\ln(\cos x + \sin x) = \ln\left(\sqrt{2}\cos t\right) = \tfrac{1}{2}\ln 2 + \ln(\cos t),$$

which is valid because $\cos t > 0$ on $[0, \frac{1}{4}\pi]$. Therefore

$$\begin{aligned} I &= \int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\left(\tfrac{1}{2}\ln 2 + \ln(\cos t)\right) \mathrm{d}t \\ &= \tfrac{1}{2}\ln 2 \int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\, \mathrm{d}t + I_2 - I_1. \end{aligned}$$

The remaining integral vanishes:

$$\int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\,\mathrm{d}t = \left[\tfrac{1}{2}\sin 2t + \tfrac{1}{2}\cos 2t\right]_0^{\frac{1}{4}\pi} = \tfrac{1}{2} - \tfrac{1}{2} = 0.$$

So, using the two results above,

$$I = I_2 - I_1 = \tfrac{1}{8}(\pi - 2\ln 2 - 2) - \tfrac{1}{8}(2\ln 2 - 2) = \frac{\pi - 4\ln 2}{8} = \frac{\pi}{8} - \frac{1}{2}\ln 2.$$

Checks

Simpson's rule gives $I_1 \approx -0.076713$, $I_2 \approx -0.030588$ and $I \approx 0.046125$, matching $\frac{1}{4}(\ln 2 - 1)$, $\frac{1}{8}(\pi - \ln 4 - 2)$ and $\frac{1}{8}(\pi - 4\ln 2)$ to ten decimal places. A quick sign check on the first integral: on $(0, \frac{1}{4}\pi)$ we have $\sin 2x > 0$ and $\ln(\cos x) < 0$, so I_1 must be negative, and indeed $\ln 2 - 1 < 0$.

Common slips

- Forgetting the boundary term in the integration by parts for I_2. It is not zero at $x = \frac{1}{4}\pi$.
- Getting the signs of $\cos(2t + \frac{1}{2}\pi)$ and $\sin(2t + \frac{1}{2}\pi)$ the wrong way round, which gives $I_1 - I_2$ or $I_1 + I_2$ instead of $I_2 - I_1$.
- Dropping the constant $\frac{1}{2}\ln 2$ from $\ln\sqrt{2}$ without checking that its coefficient integrates to zero.

Beyond STEP

Integrals of $\ln(\cos x)$ and $\ln(\sin x)$ are classics: the same symmetry tricks give the famous result $\int_0^{\pi/2} \ln(\sin x)\,\mathrm{d}x = -\frac{1}{2}\pi\ln 2$, and over a general range they lead to the Clausen function and Catalan's constant. In a first-year analysis course you'd also check that such integrals are well defined when the logarithm blows up at an endpoint, which makes them improper integrals. For lots more practice in this style, see Siklos, *Advanced Problems in Mathematics*.

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